

Hyperthermia (Environmental) / Heat Emergencies

Minor Heat Related Illnesses

- Heat edema
- Heat rash
- Heat cramps / tetany
- Heat syncope
- Heat exhaustion – weakness, headache, dizziness, chest pain, shortness of breath, nausea, vomiting, abdominal pain

Heat Stroke – A Life-Threatening Emergency

- Altered level of consciousness
- Elevated body temperature
 - Body temperature is usually higher than 41.1° C (106° F) BUT if patient has other symptoms, do not exclude heat stroke, even if temperature < 41.1° C (106° F)
- Dry skin (lack of sweating) or profuse sweating
 - Not all patients with heat stroke have anhidrosis (lack of sweating). **Do not use the lack of anhidrosis to exclude heat stroke.**
- Patients often have signs of heat exhaustion and heat syncope before developing heat stroke

Persons at High Risk for Heat Stroke – infants, elderly, athletes, outdoor laborers, alcoholics/ substance abusers, homeless, **firefighters**, obese, use of certain medications (see list)

EMT/AEMT

- Follow General Medical Care Guideline
- Move to cool environment

Altered Mental Status OR Temp > 102° F OR Syncope

- Follow Adult Resuscitation Guideline, as indicated
- Active cooling measures
 - Remove clothes
 - Mist with water
 - Fan
 - Ice packs and/or wet cloth to groin, neck, axilla
- Check temperature every 5 minutes, stop active cooling when temperature below 102° F

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO if unable to tolerate fluid by mouth
- **Zofran** 4mg IM/ODT for nausea/vomiting every 5 min x 2 doses (max 8mg)

Paramedic

Altered Mental Status OR Temp > 102° F OR Syncope

- Manage airway
- **NS/LR** 1000 mL IO/IV
 - Watch closely for pulmonary edema
- Stop cooling if shivering noted or when patient's internal body temperature reaches 39°F or external body temperature reaches 38°F
- Follow Seizure guideline if seizing noted - Continue **NS/LR** infusion in the setting of seizures

Table 2 Medications and drugs that may contribute to heat illness

Alcohol
Alpha adrenergics
Amphetamines
Anticholinergics
Antihistamines
Antipsychotics
Benzodiazepines
Beta blockers
Calcium channel blockers
Clopidogrel
Cocaine
Diuretics
Laxatives
Neuroleptics
Phenothiazines
Thyroid agonists
Tricyclic antidepressants